Internship Report on

# VLSI Physical Design

Submitted by:
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Duration: [09-JUL-2025 – 06-SEP-2025]

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MAVEN SILICON
Centre of Excellence in Semicon

**TO WHOMSOEVER IT MAY CONCERN**

This is to inform you that Mr.Prasanna Kumar Jakki has been undergoing his Internship program in the **VLSI Physical Design Internship** at Maven Silicon Pvt. Ltd.

Duration: 09th July 2025 to 6th Sep 2025.

We wish him all the best in his future endeavors.

For Maven Silicon Softech Pvt Ltd.

MAVEN SILICON SOFTECH PVT. LTD. BANGALORE

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Head- Technical Training

Place: Bengaluru
Date: 30/Aug/2025

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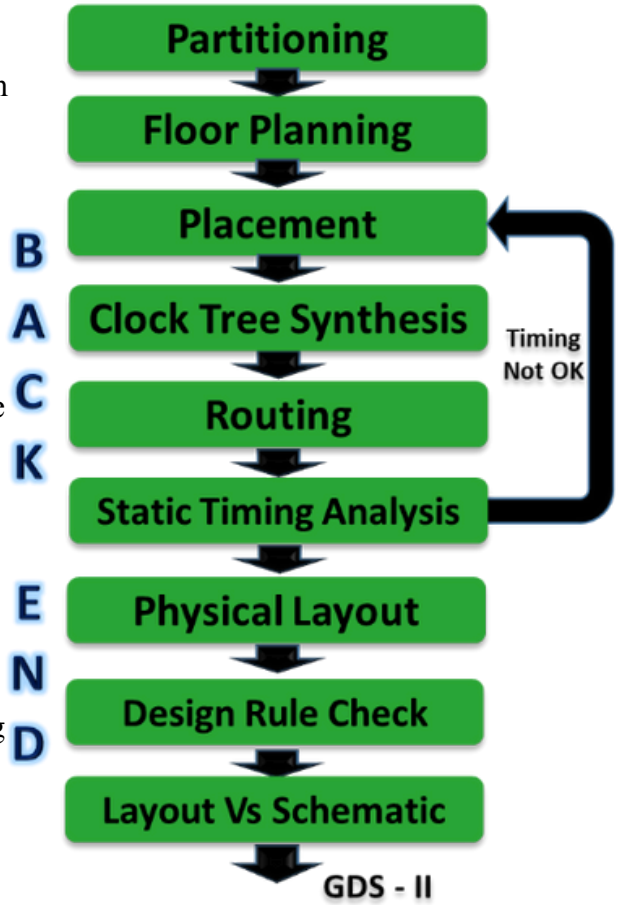
## 1. Foundation of Physical Design

Physical Design (PD) is the process of converting a gate-level netlist into a detailed, manufacturable layout file (GDSII), bridging the logical design and actual silicon implementation. This phase is crucial for meeting the goals of Power, Performance, and Area (PPA), while adhering to foundry manufacturing rules and sign-off requirements.

Physical Design Flow

- Library & Tech Imports: Load necessary libraries (LEF/DEF), technology files (.lib, tech files), and design constraints (SDC). Set up units, grid dimensions, and preferred routing directions to ensure all subsequent steps align with technology and foundry specs.
- Floorplanning: Define the die/core, allocate space for macros, plan IO pin placements, and build the power distribution network (grids, rings, straps). Budget room for clock and signal routes to prevent congestion later.
- Placement: Arrange and legalize standard cells throughout the core, optimizing for timing and congestion. Follow rules for keepouts, blockages, and halos around sensitive structures.
- Clock Tree Synthesis (CTS): Insert clock buffers/inverters, balance clock skew and latency, provide shielding for clock routes, and ensure insertion delays meet timing targets.
- Routing: Perform global followed by detailed routing across preferred metal layers, resolving shorts, spacing issues, and antenna effects.
- Post-route Optimization: Address setup/hold violations, resize cells, insert repeaters, adjust useful skew, and perform Engineering Change Orders (ECOs) to achieve final timing closure.




Key Files & Views

- LEF/DEF: Define library and design attributes for layout and placement.
- Liberty (.lib): Provide timing and power details for each cell.
- GDSII/OASIS: Final physical layout formats for mask generation.
- SPEF: Carries parasitic data for accurate timing analysis.
- SDC: Specifies all timing constraints for sign-off.

Typical Challenges

- Achieving timing closure across multiple operational modes and process corners (MMMC).
- IR drop and electromigration management to ensure power integrity and reliability.
- Mitigating routing congestion to avoid localized performance issues.
- Ensuring sign-off correlation between implementation tools and timing/parasitic extraction analyses.

## 2. Foundation of Physical Verification

Physical Verification is a critical stage in semiconductor design, ensuring that the chip layout is both manufacturable and functionally equivalent to the intended circuit before it goes to fabrication. This process helps prevent costly respins by catching physical and electrical errors preemptively.

Major Components of Physical Verification

Design Rule Check (DRC)

- DRC validates that the design conforms to foundry-specified geometric rules, such as minimum width, minimum spacing, via placements, enclosures, antenna rules, notch and jog checks, and density rules.
- Any violations can cause yield loss, shorts, or opens, potentially resulting in non-functional chips or reduced reliability.

Layout Versus Schematic (LVS)

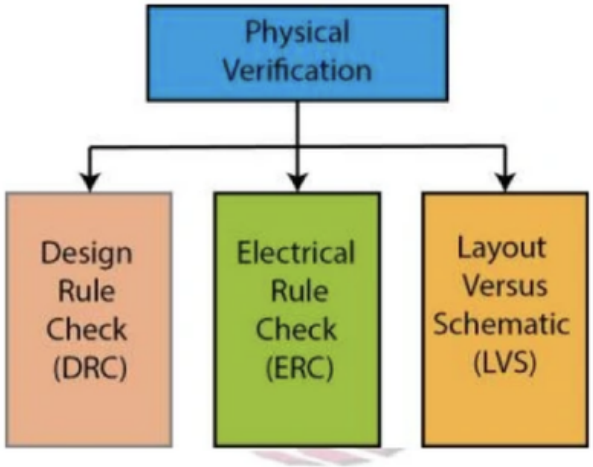
- LVS compares the netlist extracted from the final layout against the original schematic or reference netlist.
- It helps identify missing devices, unintended connections, shorts, opens, and ensures the fabricated device matches the intended logical design.

Electrical Rule Check (ERC)

- ERC verifies the electrical correctness by flagging issues like floating nodes, missing substrate or well connections, power/ground errors, over-voltage conditions, and illegal gate connections.
- These checks prevent functional problems such as excessive leakage, damaging current paths, or device malfunction.

Sign-off and Debug

- Industry-standard sign-off tools include Calibre and Pegasus.
- Debugging typically involves cross-probing layout shapes, reviewing rule decks, and querying violators. Waivers may be applied, but always require explicit consent from the foundry to avoid manufacturing risks.
- Additional checks like density fill and DFM (Design for Manufacturability) are often included to ensure the design meets chemical-mechanical planarization (CMP) and lithography constraints for robust fabrication.



## 3. VLSI Physical Design Floorplanning

Floorplanning is a foundational step in physical design that determines the layout structure and resource distribution for an integrated circuit. The choices made during floorplanning have critical impacts on timing, routability, and power integrity (IR/EM).

Die/Core Definition

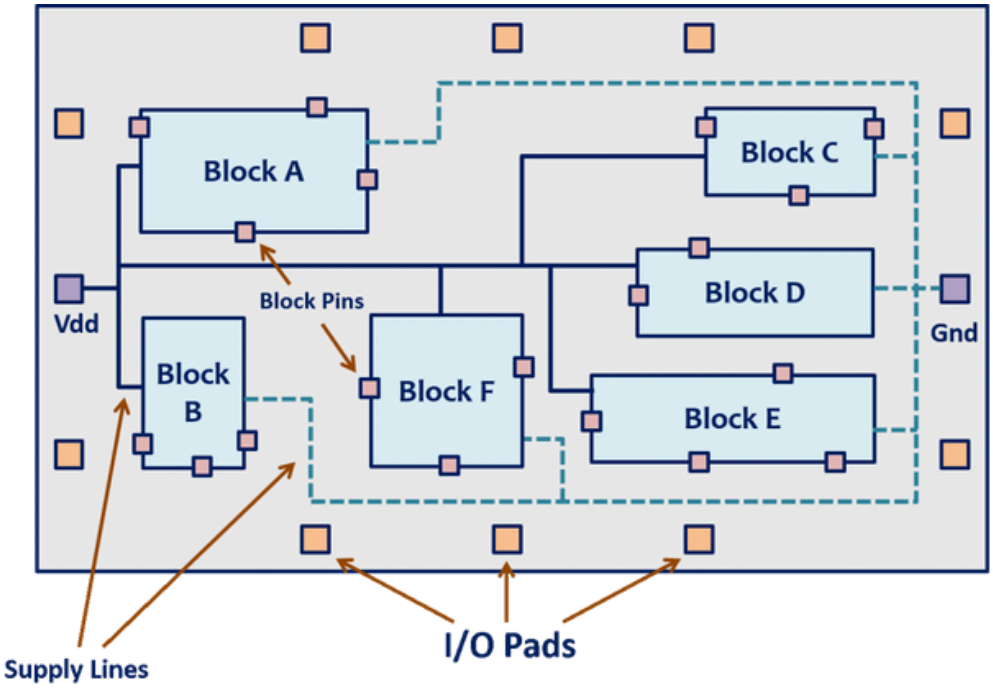
- The die outline and core area are established, setting the aspect ratio and determining utilization targets, typically between 60–75% for standard cells to balance routability and density.
- Sufficient keep-out margins and channel reservations are planned around macros, preventing routing blockages and allowing for buffers and clock tree synthesis.

Macro & IO Planning

- Large IP blocks and memories are placed first—ideally near associated IO pads—to minimize the length and delay of critical nets.
- Macros are oriented to expose sides with the most pins toward adjacent logic, simplifying routing.
- Halos (reserved buffer areas around macros) are added to provide space for routing and clock tree insertion.

Power Planning

- The power grid consists of rings, meshes, and straps usually routed on top metal layers and sized according to current density requirements.
- Via ladders are used for robust vertical current flow.
- Separate power rails should support different domains—analog, digital, always-on—to isolate noise and optimize reliability.
- Decoupling capacitors (decap insertion) are included according to guidelines for improved noise immunity.




Routability & Timing Considerations

- Detours are minimized to avoid signal delay and routing congestion.
- Space is specifically budgeted for clock spines and shields for critical timing paths.
- Timing-sensitive logic is placed close together to reduce resistance and capacitance (RC delay).

Common Pitfalls

- Excessive macro clustering can cause severe routing congestion and timing issues.
- Insufficient power grid width leads to excessive IR drop, risking unreliable operation.
- Inadequate keep-out zones around analog or sensitive IP blocks can introduce signal integrity problems.

## 4. VLSI Physical Design Placement

Placement in VLSI physical design is the step where millions of standard cells are assigned legal physical locations on the chip, optimizing for timing, congestion, and power efficiency. This process involves multiple stages, from an initial coarse global placement to a finer detailed placement with iterative optimizations.

Objectives & Metrics

- Wirelength (Half-Perimeter Wire Length - HPWL): Minimizing wirelength reduces delay and power.
- Congestion Maps: Identify and alleviate routing resource hotspots.
- Timing Metrics (WNS - Worst Negative Slack, TNS - Total Negative Slack): Guide placement improvements to meet timing.
- Power Hotspots: Areas with high activity or density are optimized to balance power use.
- Critical timing paths are often clustered or buffered to improve delay and robustness.

Techniques

- Use of analytical solvers and force-directed models mathematically optimize cell locations.
- Partitioning divides the design into manageable segments.
- Legalization resolves placement conflicts, ensuring cells do not overlap.
- Spreading moves cells out of congested bins to relieve routing pressure.

Constraints & Blockages

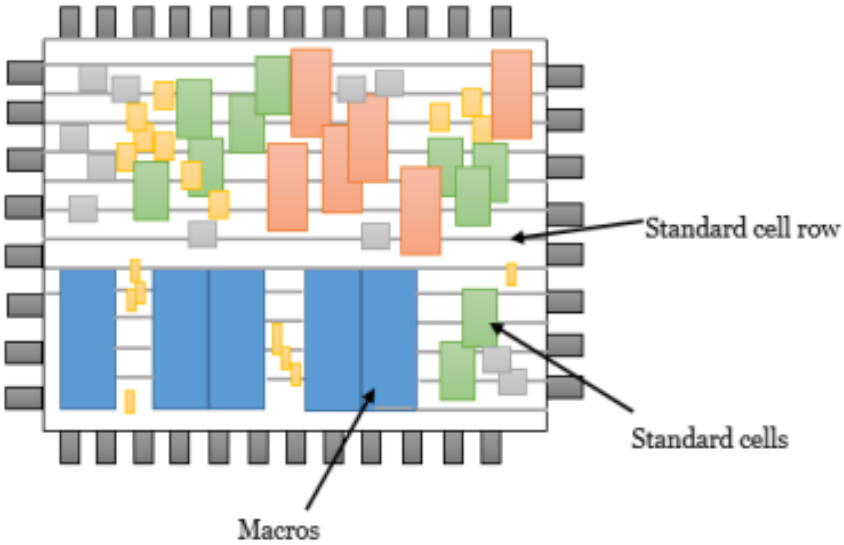
- Placement blockages and keepouts (soft or hard) restrict cell placement in sensitive or reserved areas.
- Cells near high-fanout nets are padded to reduce loading and timing issues.
- Limits on max transition and max capacitance control signal integrity and delay.

Clock-Aware Placement

- Reserve sites for clock tree synthesis (CTS) buffers.
- Align flip-flop rows to balance clock skew across the design.
- Minimize distances for clock crossings between clock domains to reduce latency variation.

Quality Checks

- Evaluate placement density to avoid overcrowding.
- Use congestion heatmaps for routing feasibility.
- Review critical path maps for timing closures.
- Check for violations of max transition and max capacitance constraints.
- Ensure proper use of tie-cells to prevent floating inputs.



## 5. VLSI Physical Design Routing

Routing in VLSI physical design transforms the netlist's logical connections into actual physical wires on layered metal stacks, ensuring compliance with design rules, timing integrity, and signal reliability while minimizing violations and noise.

Routing Flow

- Global Routing: Estimates approximate wire paths and guides congestion-aware distribution but does not fix wires to tracks.
- Detailed Routing: Assigns precise wire segments, places vias, and resolves conflicts. It ensures all nets are fully connected and all design rules are met.

Signal Integrity & Reliability

- Crosstalk is mitigated using increased wire spacing, shielding with ground lines, and changing routing layers for adjacent nets.
- Electromigration protection involves following current density limits, widening power and clock wires, and using redundant vias for reliability.

Special Nets

- Clock and Power Routes: Use the widest available metals and shielding techniques. Critical clocks are often routed with high-quality shielding to avoid noise.
- Differential Pairs: These require strict length matching and controlled coupling between wires for signal fidelity.

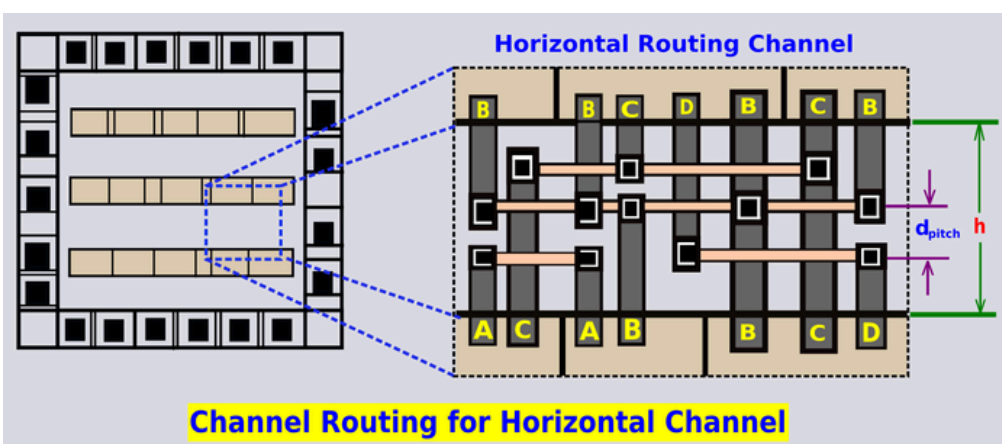
Antenna & DRC Fixes

- Antenna Effects: Resolved by adding diodes or breaking long wires into jogs.
- DRC Violations: Shorts, incorrect spacing, notches, and patterning errors are detected and corrected.
- Double Patterning: Enforced as needed for modern technology nodes.

Post-Route Optimization

- RC Extraction (SPEF): Parasitic extraction is used for post-route timing analysis, feeding the sign-off STA.
- ECO (Engineering Change Order) Buffering/Sizing: Final cell sizing and buffer insertion are adjusted to close timing and clean up last-minute DRCs.

The provided figure visually demonstrates interconnecting wires among functional blocks, showing power and ground connections alongside signal nets to highlight routing complexity.



## 6. Static Timing Analysis (STA)

Static Timing Analysis (STA) is a method used in chip design to verify that all timing paths meet required constraints under various conditions without running dynamic simulations. It calculates the worst and best delays based on cell timing data and parasitics extracted from the physical layout.

Key Concepts

- Setup and hold checks are performed on sequential elements' flop-to-flop paths and on input/output paths.
- Timing considerations include clock latency, clock skew, and jitter.
- Supports generated clocks and virtual clocks for complex designs.

Constraints (SDC)

- Common commands include create_clock, set_input_delay, set_output_delay, set_clock_uncertainty, set_false_path, and set_multicycle_path.
- Proper constraint management is critical to avoid over-constraining or under-constraining the design, which could mask timing issues or create unnecessary design fixes.

MMMC & OCV

- STA supports analysis across Multiple Modes, Multiple Corners (MMMC) including different operating modes like functional, scan, or low power, and process corners such as SS (slow-slow), TT (typical-typical), and FF (fast-fast).
- On-Chip Variation (OCV) and static timing derates like AOCV (Advanced OCV) or POCV (Parametric OCV) are applied to model variations due to manufacturing and environmental effects.

Debug & Closure

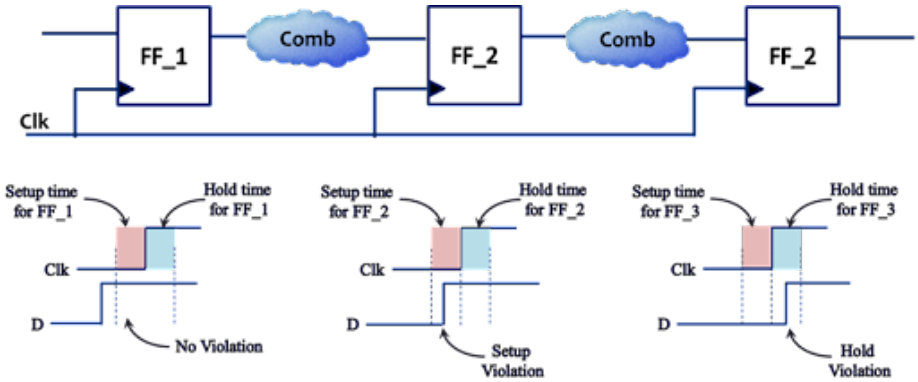
- For fixing setup violations, common methods are upsizing drivers, inserting buffers, reducing net resistance and capacitance, or applying useful clock skew.
- For hold violations, adding delay cells or rerouting nets to create detours can help.
- All changes should be re-verified for power, noise, and overall impact.

Reports

- Important timing metrics include:
  - WNS (Worst Negative Slack): Most critical timing violation.
  - TNS (Total Negative Slack): Sum of all negative slacks, indicating overall timing health.
- Reports provide path-based analysis versus graph-based and generate ECO (Engineering Change Order) logs for iterative fix application.

STA is a key verification step that ensures a reliable design under all expected variations and operational conditions.

Figure :



## 7. Foundation of Design for Testability (DFT)

DFT enhances testability by making internal nodes in a circuit controllable and observable during manufacturing tests. This approach helps achieve high fault coverage while keeping tester time reasonable.

Scan Architecture

- Flip-flops (flops) in the design are replaced with scan flops.
- These scan flops are connected into scan chains that can be shifted in and out.
- Scan enable signals and special test clocks are used to control scan operation.
- Chain lengths are planned to balance between shift time and routing overhead to optimize test efficiency.

ATPG (Automatic Test Pattern Generation) & Fault Models

- Various fault models are considered such as stuck-at faults, transition delay faults, bridging faults, and path delay faults.
- The goal is to generate a compact and effective set of test patterns that provide high fault coverage.
- Techniques like X-masking and handling test constraints are important for realistic tests.

BIST (Built-in Self-Test) & Boundary Scan

- LBIST (Logic BIST) is used for logic block self-testing.
- MBIST (Memory BIST) targets memory blocks.
- Boundary scan, standardized as JTAG (IEEE 1149.1), supports board-level testing and enables access to registers through a TAP controller.

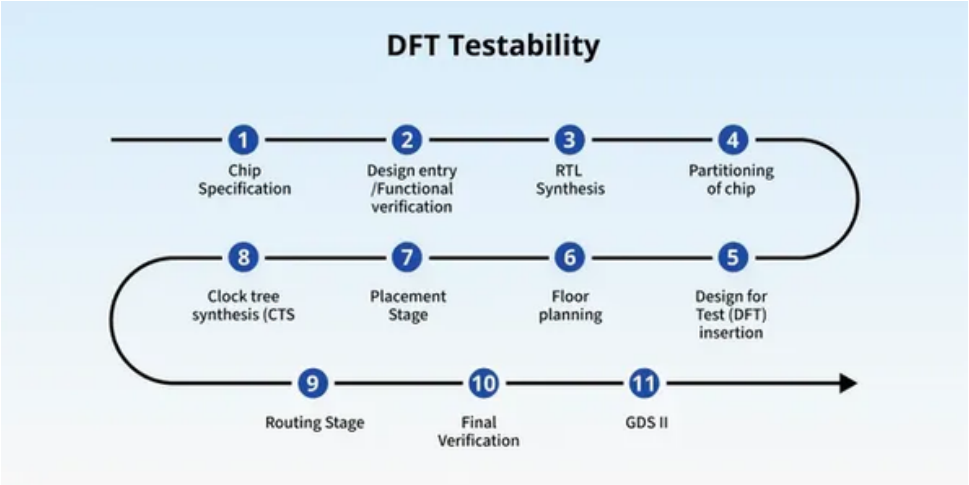
Physical Considerations

- Reserve routing space for scan chains avoiding long serpentine routes that increase delay.
- Ensure test clocks are isolated to avoid interference.
- Guarantee scan hold times are met for stablity.
- Add test points if fault coverage is insufficient or testing stalls.

Metrics & Sign-off

- Test coverage percentages, pattern counts, and test time estimates are reported.
- Verify that test modes comply with electromagnetic interference (EMI) and infrared (IR) constraints as well as input/output (I/O) specifications.

Figure :



## 8. Conclusion

This internshipdeepened my understanding of the end-to-end physical implementation flow and its tight coupling with verification, timing, and test. I practiced translating abstract constraints into tangible layout decisions, debugging timing and DRC issues, and planning for reliable power and testability. The exposure to MMMC STA, routability analysis, and sign-off verification prepares me to contribute effectively as a Physical Design or DFT engineer on real ASIC projects.